

How Application-Specific DEVELOPMENT BOARDS Help Designers



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Using a development board to design a device makes the job simpler. While selecting the board, go for one that requires minimum changes for your application. However, for intense applications, something a little more specific would be a better solution.

Development boards have become extremely useful for designers. Nitesh Kumar Jangir, founder, COEO Labs, and Vijay Bharath S., associate vice president, hardware design, Mistral Solutions, agree that use of development boards during prototyping stage of a product is of huge help.

Developers often go for custom boards to develop solutions for their specific uses, or commercial purposes, after testing these out on a regular reference board. Let us take a look at some of the solutions that could come handy the next time you decide to design a product.

Why application focus

Once you decide to go with a special board, you look into what it offers. On close

inspection, we can see some stark differences. "Reference designs are very different," says Henrik Flodell, senior product marketing manager, development tools, Microchip Technology Inc. "The goal here is to deliver something that is 80 per cent to 90 per cent complete."

The board for wearable design would be small in size, without compromising on performance in order to fit maximum peripherals onto the board. But the board for audio-video processing would have heavy data-processing capabilities. Similarly, the board focused towards the Internet of Things (IoT) application development would focus on connectivity and would be very power-conscious.

Suppose, you went with a general-purpose board for your product. You would have to add several shields on top of it to improve performance and to incorporate some features. "This can be overcome with application-focused development boards," says Hrishikesh Kamat, chief executive officer, Shalaka Technologies.

"There are highly specialised boards with specific applications or use cases in mind," adds Flodell. Kamat explains, "Now, we do not need to add shields and can design more elegant aesthetics as well."

If the final solution can be developed without going through the hassle of a separate board, this could be a welcome change. "This would also bring down the bill of materials

